EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Mid Term Examination, Spring 2025 Semester

Course:	**CSE325 - Operating Systems, Section 3**
Instructor:	**Shatabdi Roy Moon, Lecturer, Department of CSE, EWU.**
Full Marks:	**25**
Time:	**1 Hour and 15 Minutes**

Note: There are SIX questions; answer ALL of them. The course outcome (CO), cognitive level, and mark of each question are mentioned at the right margin.

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1. In an operating system, there are **four network tasks** that need to be processed by the CPU: **T1**, **T2**, **T3**, and **T4** parallelly. Each task has a processing time requirement that consists of multiple steps: **Data Fetching (F)**, **Data Processing (P)**, and **Data Saving (S)**. [CO1,C3, Mark: 5]

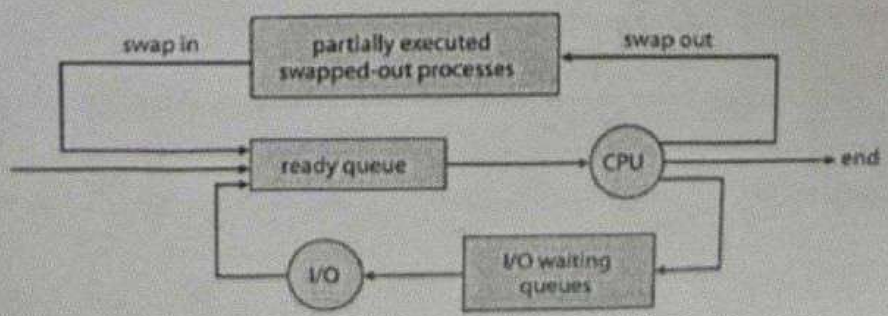
   The CPU allocates a **time slice of 40 milliseconds** to each task, and the tasks must complete each step in order (F → P → S). The CPU begins with **Task T1** at **time 0** .

   - Explain how the CPU will execute the tasks over the first **240 milliseconds**. What will the output of the tasks look like?

2. A company is designing a new **distributed computing system** for processing large amounts of data. The system will consist of multiple processors to handle the increasing workload efficiently. The system will use either **Symmetric Multiprocessing** or **Asymmetric Multiprocessing** architectures, depending on the specific use case. [CO1,C3, Mark: 5]

   - Design and construct both symmetric and asymmetric architecture for the company and analyze which model works better for the company.

3. Explain the working procedure of the scheduler as depicted in the diagram. [CO1,C3, Mark: 2]



4. Consider a multicore system and a multithreaded program written using the many-to-many threading model. Let the number of user-level threads in the program be greater than the number of processing cores in the system. Discuss the performance implications of the following scenarios. [CO2,C3, Mark: 5]

   a. The number of kernel threads allocated to the program is less than the number of processing cores.

   b. The number of kernel threads allocated to the program is equal to the number of processing cores.

   c. The number of kernel threads allocated to the program is greater than the number of processing cores but less than the number of user-level threads.

5. The following processes are being scheduled using **Shortest Remaining Time First (SRTF)** scheduling (preemptive) and **Shortest Job First** scheduling algorithm. [CO2,C4, Mark: 5]

Process ID	**Burst Time**	**Arrival Time**
P_1	18	0
P_2	20	10
P_3	15	20
P_4	12	40
P_5	10	70
P_6	8	75

For both scheduling algorithm:

   a. Show the scheduling order of the processes using a Gantt chart.

   b. Calculate the average turnaround time and average waiting time, also determine which algorithm works better.

6. In this system, each process is assigned a priority value, with **lower values indicating higher priority**. The following processes are being scheduled using **preemptive Priority Scheduling** algorithm. [CO1,C3, Mark: 3]

Process ID	**Burst Time**	**Arrival Time**	**Priority**
P_1	8	0	3
P_2	6	2	1
P_3	3	4	4
P_4	5	6	2

Build a graphical representation of the schedule of tasks that illustrate the execution of these processes by using preemptive priority scheduling algorithm.